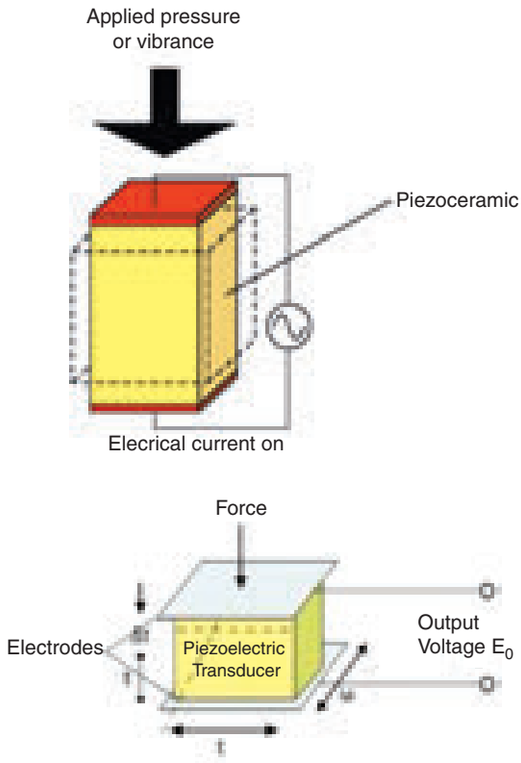


Fig. 7.13 **Piezo transducer**

a. Advantages

i. Piezo electric transducers are small in size, light in weight.
ii. It is rugged in construction.
iii. They are self-generating type of transducers.
iv. Good frequency response.
v. Measurable output.

b. Disadvantages

i. Output voltage is being affected by temperature.
ii. Leakage resistance affects its performance.

7.7 Thermocouples

In 1821, According to Thomas Seebeck, "When two dissimilar metals are in contact, a voltage is generated where the voltage is a function of temperature". This is the basic working principle of thermocouples.

This principle is used to convert heat energy into electrical energy at the junction of two conductors. This thermocouple works on principle of thermo-electric effect and this thermo-electric emf depends on the difference in temperature between the hot junction and reference junction. The thermocouples are placed inside protective wells without interruption of the working environment.

As an example, joining copper and constantan produces a voltage in the order of milli volts with the positive terminal as copper. An increase in temperature causes an increase in voltage.

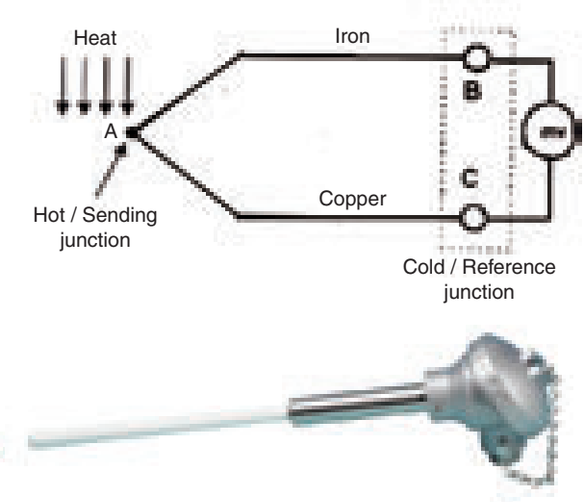


Fig. 7.14 **Thermocouples**

a. Advantages

i. Self-powered transducers.
ii. Simple and rugged in construction.
iii. Wide range of temperatures measurements.
iv. Inexpensive.

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